SANJAY SUBRAMANYAN

sanjaysubramanyan1324@gmail.com | github.com/Cosmic-dorito | linkedin.com/in/sanjay-subramanyan | +91 98408 53213 | Coimbatore, Tamil Nadu

EDUCATION

2023–2027	**B.Tech, Electronics and Communication Engineering (AI/ML Minor)**	
	Amrita Vishwa Vidyapeetham, Coimbatore	Core CGPA: **8.33** | Minor CGPA: **9.26**
2023	Class XII, Chennai Public School, Anna Nagar	**89%**
2021	Class X, Chinmaya Vidyalaya, Virugambakkam	**93.2%**

EXPERIENCE

Research Intern, National Institute of Ocean Technology (NIOT), Chennai *Apr 2026*

- Built a hardware-software co-design for a prototype **ROV-UAV hybrid**: integrated Blue Robotics **T200 thrusters** + Basic ESCs (PWM, Raspberry Pi 5), **Xsens MTi-630 INS** (quaternion, Delta-V strapdown), and **Bar30 depth sensor** in a pressure housing rated to **600 m**; implemented dual-PID control (heading + depth, 50 Hz) with INS dead-reckoning and a JSON/UDP waypoint protocol.
- Designed an **acoustic underwater communication** sub-system (FSK with piezoelectric transducers); independently designed and tested an **optical LED link** with triple-redundancy encoding (1-byte → 3-byte majority vote) and ARQ retransmission; loop-back testing confirmed error-free delivery.

AI/ML Intern, TVS Motor Company Ltd., Hosur *May – Jun 2026*

- Developed an **LLM-based patent intelligence extraction pipeline** using Azure OpenAI (GPT-4o-mini); two-layer architecture combining deterministic regex parsing for structured cover-page metadata and grounded LLM reasoning for engineering intelligence fields (technical domain, novelty, prior-art landscape).
- Built a **Streamlit** web application supporting batch patent PDF ingestion and structured **14-column Excel report** generation, with per-field confidence scores and full-document **OCR arbitration** (Tesseract) recovering scanned and design patents; deployed internally for TVSM R&D teams.
- Designed independent claim decomposition logic with TVSM design-guideline cross-mapping; validated end-to-end against real Indian patent filings.

Engineering Intern, Hyundai Motor India Limited *May – Jun 2025*

- Evaluated automated assembly workflows for **HS220** and **HS160L** six-axis robotic arms; analysed cycle-time distributions, motion-sequencing logic, and throughput constraints; mapped kinematic envelopes, workspace limits, and collision-avoidance zones.

PROJECTS

Adaptive Compute Orchestration for a Heterogeneous Edge Cluster *Ongoing*

- Built an **SBC cluster** on a dedicated Gigabit data plane (**942 Mbit/s**, 0.31 ms; K3s, Prometheus/Grafana; HPL **10.20** → **20.96 GFLOPS**, 2.05×) with an **adaptive scheduler** partitioning distributed inference by online-learned node capability, queue backlog, communication cost and thermal derating.
- Incorporating **bandwidth/latency constraints** into partitioning decisions (vs. high-speed-interconnect assumptions); on a 210-task MobileNet-v2 stream: **105/105** split (load-balance efficiency **1.000**), ∼**74 μs** scheduler overhead per task; extending to Pi 3 nodes for heterogeneity and thermal-stress evaluation.

FPGA Brain Training Game — RTL Design, Synthesis & Hardware Deployment on Basys-3

- Designed and synthesised a real-time memory game in **Verilog** on Basys-3 (XC7A35T-1CPG236C); modular RTL: **FSM**-based game logic, LFSR sequence generator, 4×4 keypad controller with debounce, clock divider (100 MHz → 1 kHz / 100 Hz / 2 Hz), and 7-segment display driver; verified in **ModelSim** before deployment.
- Post-synthesis Vivado report: **0.56% flip-flop utilization**, **0.193 W** on-chip power; deployed to physical Basys-3 board with PMOD KYPD and IR sensor.

Physics-Informed Thermal Field Reconstruction & Inverse Block-Power Recovery

- **Forward** (95 W 6-block chip, validated on 19-block EV6 geometry): benchmarked PINN vs. physics-mean **GP data assimilation** — GP **0.21–0.57 K** vs. PINN 0.42–0.60 K MAE at 2–3 orders lower compute; under nonlinear leakage (8 sensors) the PINN holds **0.82–1.09 K** vs. 1.34 K and recovers the leakage coefficient.
- **Inverse (Green's basis):** self-consistent Green's-basis PINN with trainable per-block powers fixed spectral collapse on EV6 (8.04 K → **1.01 K**); training-free **D-optimal sensor placement** reaches **0.60 K**.

Bearing Fault Diagnosis via Multi-Domain Deep Learning Fusion

- Dual-stream **1-D Time-CNN** + **FD-MRA** (RoPE self-attention, trainable frequency masks) fused via 256-D bottleneck; **0.8 ms** post-bottleneck inference; achieved **99.82% accuracy**, macro F1 **0.9982** across 4,949 hold-out samples, four fault classes, 0–3 HP variable load.

TECHNICAL SKILLS

HDL / RTL	Verilog, Vivado, ModelSim
Embedded	C/C++, Assembly, Arduino, Raspberry Pi
ML / Sci-ML	PyTorch, TensorFlow; CNNs, SNNs, PINNs
Software	Python, MATLAB, Streamlit; Azure OpenAI API, LLM pipelines
EDA / Simulation	ANSYS HFSS, SEMulator3D, LTSpice, EasyEDA, Keil μVision, Proteus, Cedar Logic
Languages	English (Full Professional) | Tamil (Full Professional)

CERTIFICATIONS

Jan–Apr 2026 **NPTEL: Embedded Systems Design** (IIT Kharagpur) **Elite** · **Top 1%** · Score: 75%